Evan Collins

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EDUCATION

Massachusetts Institute of Technology — 2022 – present
Doctor of Philosophy (PhD) Candidate in Biological Engineering — Cambridge, MA
Advised by Robert Langer ScD, Daniel Anderson PhD
GPA: 5.00/5.00

Yale University — 2018 – 2022
Master of Science (MS) + Bachelor of Science (BS) in Biomedical Engineering — New Haven, CT
GPA: 3.96/4.00; Highest in major; High Honors with Distinction
Tsinghua University (Summer 2020): Inter-University Program for Chinese Language Study

CORE WORK EXPERIENCE

Koch Institute for Integrative Cancer Research at MIT – Graduate Researcher — 2022 – present, Cambridge, MA
- Conducting nanomedicine research using AI+robotics to engineer improved mRNA therapeutics for vaccines and gene editing; advised by Daniel Anderson and Robert Langer

Yale School of Medicine – Student Investigator — 2018 – 2022, New Haven, CT
- Conducted systems neuroscience research for epilepsy treatment; advised by Hitten Zaveri and Dennis Spencer
- Conducted cognitive psychology research on combatting misinformation online; advised by Gordon Pennycook and David Rand

Simplex Sciences – Chief Operating Officer & Chief Scientific Officer — 2019 – 2022, New Haven, CT
- Led 10-member team to produce and sell single-stranded DNA ladders as size standards for gel electrophoresis
- Fulfilled orders across academia and biotech; products cited in top journals including *Cell* and *Nature Protocols*

D. E. Shaw & Co. – Risk Analyst Intern — 2021, New York, NY
- Developed methods to quantify market risk and evaluate portfolio-level exposures to support risk-management decisions

Click Therapeutics – Data Scientist — 2020 – 2021, New York, NY
- Built deep learning models for speech-based emotion detection to support digital therapeutics for schizophrenia

United States House of Representatives – Congressional Intern — 2017 – 2018, Middleburg, FL
- Supported constituent casework involving immigration, Veterans Affairs, and Social Security

AWARDS

- Siebel Scholar Award (2026 – 2027)
- MIT HEALS Graduate Fellowship (2026 – 2027)
- Gordon Tang Goodwin Student Support Fund Recipient (2026)
- Sigma Xi Scientific Research Honor Society (2026)
- MIT Wishnok Prize for Best Talk (2025)
- MIT Prize for Open Data Honorable Mention (2025)
- MIT Jameel Clinic Fellowship for Machine Learning in Health (2022)
- Yale D. Allan Bromley Prize for Highest GPA in Biomedical Engineering (2022)
- Yale SEAS Belle and Carl Morse Scholarship (2021)
- D. E. Shaw Nexus Fellowship (2021)
- Yale Richard U. Light Fellowship for Chinese Studies (2020)
- Tau Beta Pi Engineering Honor Society (2020 – 2022)
- Lockheed Martin STEM Scholarship (2019 – 2022)
- Yale Summer Research Fellowship (2019)
- Intel International Science and Engineering Fair Third Place Prize (2018)
- National Merit Finalist and Scholarship (2018)

- Valedictorian of Ridgeview High School (2018)
- Florida Sunshine State Scholar (2017)

TEACHING

Machine Learning for Molecular Engineering (3.C01/3.C51/10.C01/20.C01/20.C51) 2024
Massachusetts Institute of Technology Cambridge, MA
Course Content Developer

Principles of Molecular Bioengineering (20.420[J]) 2023
Massachusetts Institute of Technology Cambridge, MA
Teaching Assistant

Physiological Systems (BENG 350) 2020
Yale University New Haven, CT
Peer Tutor

Ordinary and Partial Differential Equations (ENAS 194) 2020
Yale University New Haven, CT
Peer Tutor

PUBLICATIONS

*Equal contribution, †Corresponding author

Collins E*, Aguilar-Bejarano E*, Manan R S*, Ji J, Ko G, Im W, Natarajan V, Langer R, Anderson D G† TBD
In preparation

Collins E*, Seth A*, Langer R, Anderson D G† TBD
In preparation

Collins E*, Chen M*, Chen Y*, Langer R, Anderson D G† TBD
In revision at Nature Reviews Materials

Collins E*, Manan R S*, Detmer S, Muser T, Raman E, Zhu R, Balagopal S, Shi Y, Ji J, Gupta A, Hu Y, Trongjit S, Chung W, Mughal H, Witten J, Lapteva A, Pasupathy A, Im W, Langer R, Anderson D G† TBD
In revision at Nature Biomedical Engineering

Detmer S, **Collins E**, Mughal H, Raman E, Pasupathy A, Manan R S, Balagopal S, Hu Y, Chauhan V P, McMullen P, Chen Y, Shi Y, Lu K, Self A, Hall A, Lee R, Langer R, Anderson D G† TBD
In revision at ACS Nano

Mughal H*, Self A*, Chen Y*, McMullen P, Beyer E, Manan R S, Raman E, Detmer S, Birdsall B, Chen B, Monkovic J, Balagopal S, Pasupathy A, **Collins E**, Rudra A, Langer R, Anderson D G† TBD
In revision at Nature Biotechnology

Hu Y*, Prado M*, Chen B*†, Rudra A, Reed K, Trongjit S, Bartrum O, Chen Y, Detmer S, McMullen P, **Collins E**, Chen Y, Jiang A Y, Danko A, Shi D, Byrne S, Wenning M, Prokopovych L, Langer R, Anderson D G† TBD
In revision at Nature

Zhao Y*, Jeang W J*, Xu M, Perez I, Razavi S, Carluccio G, **Collins E**, McMullen P, Kumar A, Manan R S, Li N, Detmer S, Collins J J, Langer R, Anderson D G† TBD
Under review at Science

"Quantitative Cryogenic Orbitrap Secondary Ion Mass Spectrometry (Q-Cryo-OrbiSIMS) demonstrated for lipid nanoparticles (LNPs)" TBD
Roberts J, Kotowska A, Aguilar-Bejarano E, **Collins E**, Witten J, Howard A, Trindade G, Gilmore I, Hook A, Winkler D, Alvey K, Figueredo G, Scurr D J, Anderson G D†, Alexander M R†
ChemrXiv. In revision at Advanced Materials
https://doi.org/10.26434/chemrxiv.15001097/v1

"Translating a review of language function to the parcellation of a high-resolution brain atlas" 2026
Watson E†, **Collins E**, Chishti O, Bose-Roy R, McGrath H, Segal N A, Lehman R, Sivaraju A, Zaveri H P, Spencer D D†
Epilepsia
https://doi.org/10.1002/epi.70229

"Lipid Nanoparticle Database towards structure-function modeling and data-driven design for nucleic acid delivery" 2026
Collins E*, Ji J*, Kim SG*, Witten J, Kim S, Zhu R, Park P, Jung M, Park A, Manan R S, Rudra A, Jeang W J, Langer R, Anderson D G†, Im W†
Nature Communications
https://doi.org/10.1038/s41467-026-68818-1

"Multitask Language Mapping to Visualize the Spatial Configuration of Polyfunctional Language Cortex" 2025
Chishti O, **Collins E**, McGrath H, Zhang T, Quraishi I H, Hirsch L J, Benjamin C, Damisah E C, Zaveri H P, Spencer D D, Sivaraju A†
Neurology
https://doi.org/10.1212/WNL.0000000000214256

"Self-driving labs for biotechnology" 2025
Collins E, Langer R, Anderson D G†
Nature Computational Science
https://doi.org/10.1038/s43588-025-00885-8

"Yale Brain Atlas to interactively explore multimodal structural and functional neuroimaging data" 2025
Collins E†, Chishti O, McGrath H, Obaid S, King A, Qiu E, Gabriel E, Shen X, Arora J, Papademetris X, Constable T R, Spencer D D, Zaveri H P†
Frontiers in Network Physiology
https://doi.org/10.3389/fnetp.2025.1585019

"Antifouling Immunomodulatory Copolymer Architectures that Inhibit the Fibrosis of Implants" 2025
Jeang W J*, Wong B M*, Zhao Y, Manan R, Jiang A L, Bose S, **Collins E**, McMullen P, Rosenboom J, Lathwal S, Langer R, Anderson D G†
Advanced Materials
https://doi.org/10.1002/adma.202414743

"Artificial intelligence-guided design of lipid nanoparticles for pulmonary gene therapy" 2024
Witten J*, Raji I*, Manan R S*, Beyer E, Bartlett S, Tang Y, Ebadi M, Lei J, Nguyen D, Oladimeji F, Jiang A Y, MacDonald E, Hu Y, Mughal H, Self A, **Collins E**, Yan Z, Engelhardt J F, Langer R, Anderson D G†
Nature Biotechnology
https://doi.org/10.1038/s41587-024-02490-y

"Mapping the structure-function relationship along macroscale gradients in the human brain" 2024
Collins E†, Chishti O, Obaid S, King A, McGrath H, Shen X, Arora J, Papademetris X, Constable T R, Spencer D D, Zaveri H P
Nature Communications
https://doi.org/10.1038/s41467-024-51395-6

"Systematic 1 Hz Direct Electrical Stimulation for Seizure Induction: A Reliable Method for Localizing Seizure Onset Zone and Predicting Seizure Freedom" 2024
Sivaraju A†, Quraishi I, **Collins E**, McGrath H, Ramos A, Turk-Browne N, Zaveri H P, Damisah E, Spencer D D, Hirsch L J
Brain Stimulation
https://doi.org/10.1016/j.brs.2024.03.011

"High-resolution cortical parcellation based on robust brain landmarks for precise localization of multimodal data" 2022
McGrath H†, Zaveri H P, **Collins E**, Jafar T, Chishti O, Obaid S, Ksendzovsky A, Wu K, Papademetris X, Spencer D D
Scientific Reports
https://doi.org/10.1038/s41598-022-21543-3

"The Implied Truth Effect: Attaching Warnings to a Subset of Fake News Headlines Increases Perceived Accuracy of Headlines Without Warnings" 2020
Pennycook G, Bear A, **Collins E,** & Rand D G†
Management Science
https://doi.org/10.1287/mnsc.2019.3478

CONFERENCES

Collins E, Ji J, Kim SG, Witten J, Kim S, Zhu R, Park P, Jung M, Park A, Manan R S, Rudra A, Jeang W J, Im W, Langer R, Anderson D G. "Lipid Nanoparticle Database (LNPDB) towards structure-function modeling and data-driven design for nucleic acid delivery". Poster at *ChemAI NYC at NYU Simons Center for Computational Physical Chemistry* (2026).

Collins E. "Data-driven approaches for understanding and improving mRNA lipid nanoparticles". Talk at *MIT Bioengineering & Toxicology Seminar* (2026).

Collins E. "Data-driven approaches for understanding and improving mRNA lipid nanoparticles". Talk at *MIT Koch Institute for Integrative Cancer Research Retreat* (2025).

Collins E, Ji J, Kim SG, Witten J, Kim S, Zhu R, Park P, Jung M, Park A, Manan R S, Rudra A, Jeang W J, Im W, Langer R, Anderson D G. "LNPDB: structure-function dataset of lipid nanoparticles to advance data-driven design for nucleic acid delivery". Poster at *MIT Molecular Machine Learning Conference (MoML)* (2025).

Park P, **Collins E**, Kim S, Im W. "Improving Martini 3 Coarse-Grained Models for Ionizable Lipids in LNP Simulations". Poster at *American Chemical Society Fall 2025* conference (2025).

Ji J, **Collins E**, Kim SG, Kim S, Im W. "LNPDB: structure-function database of lipid nanoparticles to advance data-driven design for nucleic acid delivery". Poster at *American Chemical Society Fall 2025* conference (2025).

Collins E. "Autonomous robotic platform for the high-throughput directed evolution of mRNA lipid nanoparticles". Talk at *MIT Bioengineering & Toxicology Seminar* (2024).

Collins E, Chen M, Langer R, Anderson D G. Poster at *MIT Department of Biological Engineering Retreat* (2024).

Watson E, **Collins E**, Chishti O, McGrath H, Sivaraju A, Zaveri H P, Spencer D D. "Yale Brain Atlas Parcellation of Overlapping Multimodal Functions in the Human Temporal Lobe". Poster at *American Epilepsy Society* conference (2024).

King A, Obaid S, Pu K, Chishti O, **Collins E**, McGrath H, King-Stephens D, Duckrow R, Spencer D D, Zaveri H P. "Use of the Yale Brain Atlas to Determine Signatures of Cortical Thinning". Poster at *American Epilepsy Society* conference (2024).

Pu K, Chen M, Chen C, King A, **Collins E,** Spencer D D, Zaveri H P. "PET Hypometabolism is Associated with Epileptogenesis at Multiple Spatial Resolutions Ranging from the Hemisphere of Seizure Onset to the Seizure Onset Area". Poster at *American Epilepsy Society* conference (2023).

Watson E, **Collins E,** Chishti O, McGrath H, Sivaraju A, Zaveri H P, Spencer D D. "The optimization of a high-resolution brain atlas as a tool for surgical planning by translating a review of language function to parcellation". Poster at *American Epilepsy Society* conference (2023).

Collins E, Chishti O, Obaid S, McGrath H, King A, Spencer D D, Zaveri H P. "Large-scale analysis of the relationship between structure and function in the human cortex". Poster at *American Society of Neuroradiology* Annual Meeting (2023).

Collins E, Chishti O, Obaid S, McGrath H, Jafar T, King A, Zaveri H P, Spencer D D. "Anatomical brain atlas for structure-function coupling: predicting surgical outcome from seizure spread and analyzing large-scale fMRI-connectome associations". Poster and presentation at *American Epilepsy Society* conference (2022).

Collins E, McGrath H, Zaveri H P, Papademetris X, Wu K, Spencer D D. "Systematic Parcellation of the Human Cortex for Comparison of Multimodal Neuroimaging Data". Poster at *American Epilepsy Society* conference (2021).

Jafar T, McGrath H, Ksendzovsky A, Zaveri H, Farooque P, **Collins E**, Sivaraju A, Damisah E, Papademetris X, Spencer D D. "A multimodal cortical atlas for clinical decision making and function-structure hypotheses in epilepsy surgery". Poster at *Society for Neuroscience* conference (2021).

ACADEMIC SERVICE

Mentorship

- Massachusetts Institute of Technology, Undergraduate thesis research — 2023 – present
- Yale University, Undergraduate thesis research — 2021 – present

Public engagement and science communication

Press interviews for *Nature, YaleNews*

Peer review

Contributed reviews to *BMC Medical Imaging, Brain Structure and Function, Brain and Cognition, Epilepsy & Behavior, Frontiers in Neuroscience, IBRO Neuroscience Reports, Molecular Therapy-Nucleic Acids, Nature Biotechnology, Nature Reviews Materials, Proceedings of the National Academy of Sciences, Scientific Reports, Technology in Society*